FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Advanced)-2019

PAPER – 1

TEST DATE: 13-05-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A, B, C, D

Sol. Total displacement is zero.

Total distance travelled = $\left(\frac{1}{2}\times 10\times 20\right)\times 2 = 200\text{m}$

2. B

Sol. Equation of motion perpendicular to inclination

$10 \sin 30° - g \cos 30° t = 0$

$t = \frac{10}{10\sqrt{3}} \Rightarrow \left(t = \frac{1}{\sqrt{3}}\right)$

Time of flight $T = 2t = \frac{2}{\sqrt{3}}$ sec

3. A, B

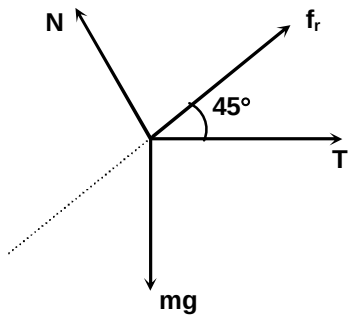
Sol. Torque about centre must be zero

$\Rightarrow T = f_r$

$T \cos 45° + f_r = mg \sin 45°$

$T\frac{\left(\sqrt{2}+1\right)}{\sqrt{2}} = \frac{100}{\sqrt{2}}$

$T = \frac{100}{\left(\sqrt{2}+1\right)}$



4. A, B, D

Sol. $F = -\frac{dU}{dx}$

5. A, D

Sol. The maximum energy of photon depends on the energy of electrons incident.

6. A, B

Sol. Optical path difference = $\left(n_1L_1 - n_2\lambda_2\right)$

$\Delta\phi = \frac{2\pi}{\lambda}\Delta x$

7. A

8. A

Sol. (for Q.7-8)

$\Delta\phi = \pi R^2\left(3B_0 - 2B_0\right) = B_0\pi R^2$

$\omega t = \frac{\pi}{2} \Rightarrow t = \frac{\pi}{2\omega}$

$\frac{\Delta\phi}{\Delta t} = \frac{2\beta_0\pi R^2\omega}{\pi}$

$\vec{\tau} = \vec{\mu}\times\vec{B}$

9. A

10. A

Sol. (for Q.9-10)

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$= \frac{1}{30} + \frac{1}{60}$

$\Rightarrow \frac{1}{f} = \frac{1}{20} \Rightarrow f = 20cm$

Focal length of convex mirror

$f = \frac{1}{2}(30 - 10) = 10cm$

SECTION – D

11. 01200.00

Sol. $V_1 = \frac{1}{4\pi\in_0}\times\frac{q}{\sqrt{x^2 + R^2}}$

$V_1 = \frac{9\times10^{+9}\times10\times10^{-9}}{3\times10^{-2}} = 3000$

$V_2 = \frac{9\times10^{9}\times10^{-9}\times10}{5\times10^{-2}} = 1800$

$\Delta V = 1200\ V$

12. 00781.25

Sol. $E_1 = \frac{r}{2}\frac{dB}{dt}$

$E_2 = \frac{R^2}{2r}.\frac{dB}{dt}$

$= \frac{3}{2}\times 5\times 10^{-2}\times\frac{25}{2\times 6}\times 5\times 10^{-2}$

$= \frac{25\times 25\times 3\times 10^{-4}}{4\times 6} = 78.125\times 10^{-4} = 781.25\times 10^{-5}$

13. 00002.88

Sol. $P = \frac{V^2}{4R} \Rightarrow R = \frac{V^2}{4P} = \frac{48\times 48}{4\times 200} = 2.88\Omega$

14. 00093.75

Sol. $I = \frac{m\ell^2}{12} = 20\times\frac{3}{4}\times\frac{3}{4}\times\frac{1}{12} = \frac{180}{192}$

15. 00010.20

Sol. Excess pressure

$\Delta P = \frac{2T}{R} = \frac{2\times 7\times 10^{-2}}{0.2\times 10^{-3}} = 700\ N/m^2$

Pressure inside bubble = $1.013\times 10^5 + 700$

$= 10.13\times 10^4 + 0.07\times 10^4 = 10.20\times 10^4\ N/m^2$

16. 00026.68

Sol. $E = \frac{GM}{2R^2} = \frac{6.67\times 10^{-11}\times 2}{2\times\left(\frac{1}{2}\right)^2} = 6.67\times 4\times 10^{-11} = 26.68\times 10^{-11}\ N/kg$

17. 00000.60

Sol. $A_t = A_i\frac{2v_2}{(v_1+v_2)} = \frac{2\times 2\times 0.9}{6} = 2\times 0.3 = 0.6$

18. 00014.00

Sol. Slope $\frac{dP}{dV} = -\frac{\gamma P}{V}$

$\frac{7}{5}\times 10^5 = V$

$V = \frac{70}{5}\times 10^4\ m^3 = 14\times 10^4\ m^3$

Chemistry

PART – II

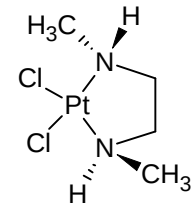
SECTION – A

19. A, C

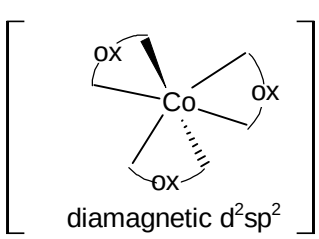
Sol. Be also reacts directly with fluorine Except Be and Mg, other alkaline earth metals give flame test.

20. A, B

Sol.



diamagnetic dsp^2



diamagnetic d^2sp^2

$K\left[Cr(H_2O)_2(C_2O_4)_2\right]$

paramagnetic

$\left[Ni(en)_2Cl_2\right]$

paramagnetic

21. B, C

Sol. $F_2 + H_2O \longrightarrow HF + O_2$

$I_2 + H_2O \longrightarrow$ No reaction

$Cl_2 + H_2O \longrightarrow HCl + HClO$

$Br_2 + H_2O \longrightarrow HBr + HBrO$

22. B, C, D

Sol. Peptization is the process of forming colloidal sol by freshly prepared precipitate in the presence of small amount of electrolyte.

23. B, C

Sol. As temperature decreases or molar mass decreases

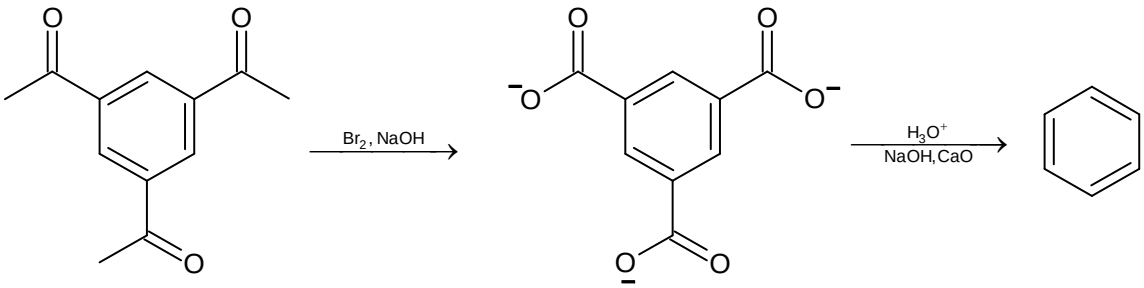
H increases

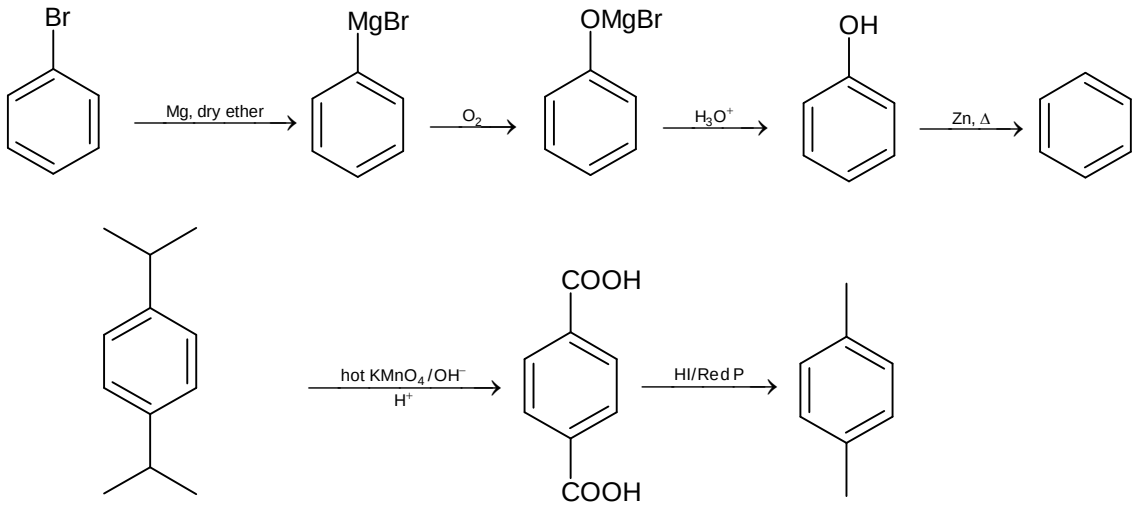
L decreases

w decreases

24. A, C, D

Sol.

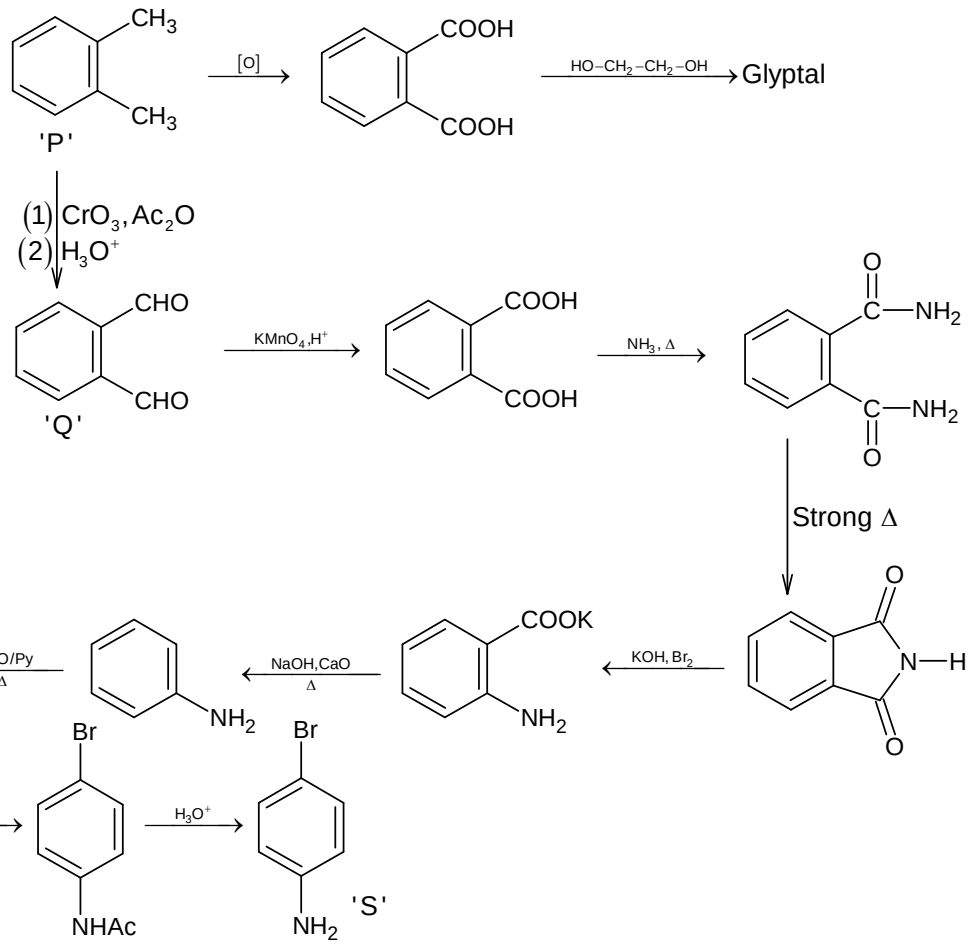


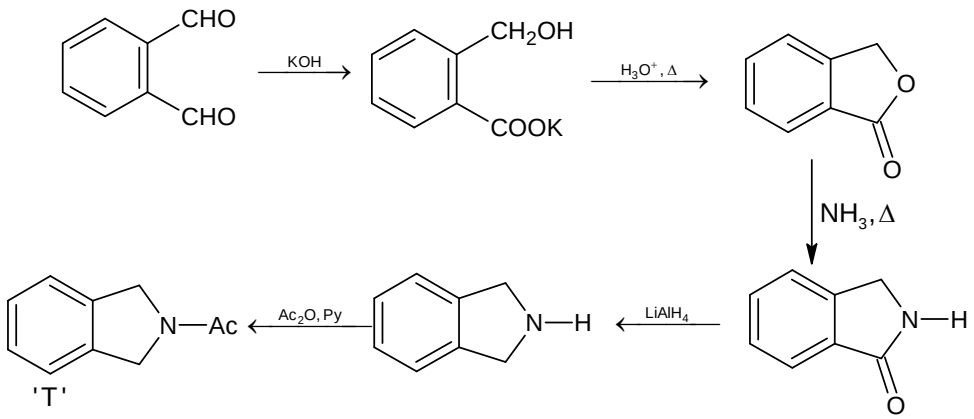

Br
Mg, dry ether
MgBr
O_2
OMgBr
H_3O^+
OH
Zn, Δ
hot $KMnO_4/OH^-$
H^+
COOH
COOH
HI/Red P

25. C

26. B

Sol.

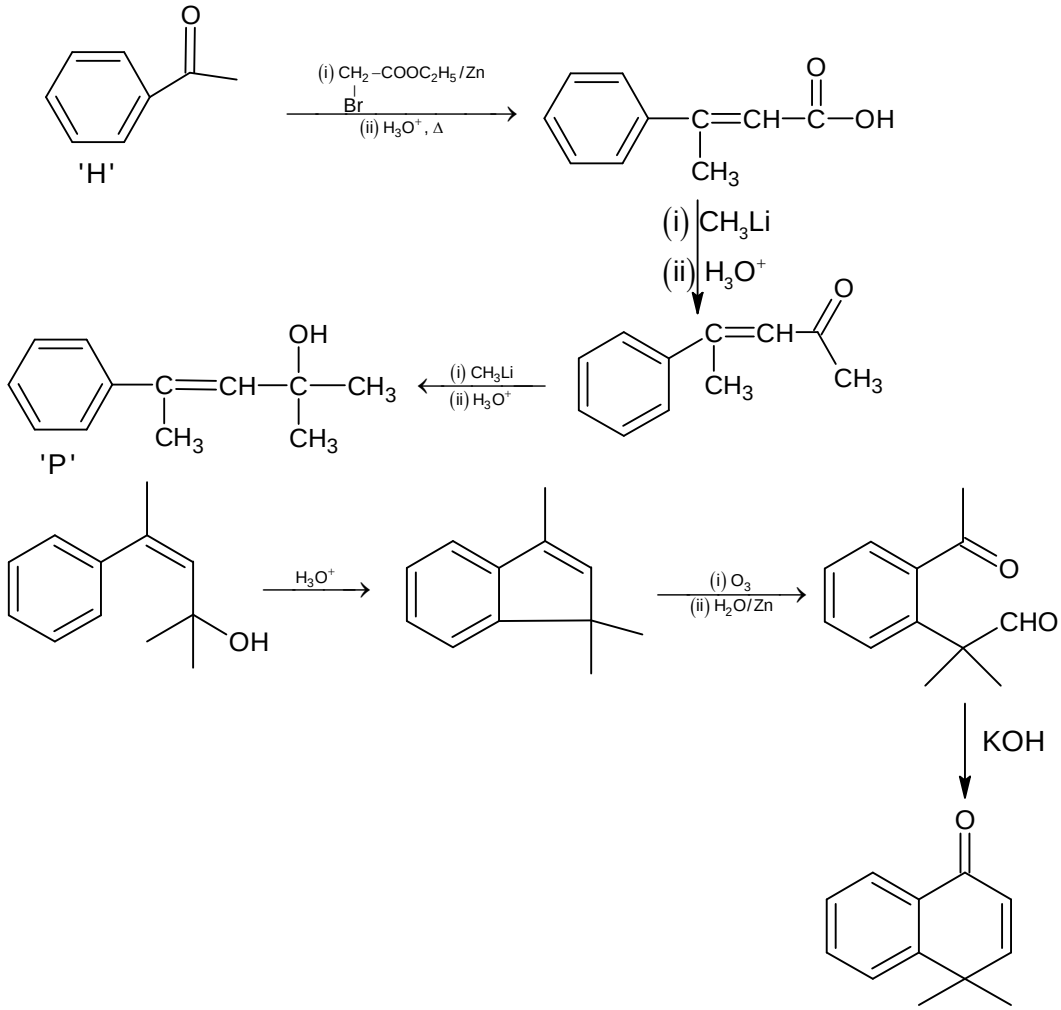

CH_3
CH_3
'P'
[O]
COOH
COOH
$HO-CH_2-CH_2-OH$
Glyptal
(1) CrO_3, Ac_2O
(2) H_3O^+
CHO
CHO
'Q'
$KMnO_4, H^+$
COOH
COOH
NH_3, Δ
NH_2
NH_2
Strong Δ
KOH, Br_2
COOK
NH_2
NaOH, CaO
Δ
NH_2
Ac_2O/Py
Δ
HN—Ac
Br_2/Ac_2O
Br
NHAc
H_3O^+
Br
NH_2
'S'


CHO
CHO
KOH
CH2OH
COOK
H3O+, Δ
O
O
NH3, Δ
N—H
O
LiAlH4
N—H
Ac2O, Py
N—Ac
'T'

27. C

28. B

Sol.


O
'H'
(i) CH2–COOC2H5/Zn
Br
(ii) H3O+, Δ
O
C=CH–C–OH
CH3
(i) CH3Li
(ii) H3O+
C=CH
O
CH3
CH3
OH
C=CH–C–CH3
CH3
CH3
(i) CH3Li
(ii) H3O+
'P'
H3O+
OH
(i) O3
(ii) H2O/Zn
O
CHO
KOH
O

SECTION – D

29. 00010.00

Sol. $d = \dfrac{Z \times m}{N_A \times a^3}$

$90 \times 10^{-3} = \dfrac{4 \times 108}{6 \times 10^{23} a^3}$

$a^3 = \dfrac{4 \times 108}{6 \times 9 \times 10^{21}}$

$a^3 = 8 \times 10^{-21}$

$a = 2 \times 10^{-7}$ cm

$a = 2 \times 10^{-9}$ m

$a = 20 \times 10^{-10}$ m

Distance between X & Y $= \dfrac{20 \times 10^{-10}}{2}$

$= 10 \times 10^{-10}$ m

30. 00051.20

Sol. $Q_P = \dfrac{(200)^2}{(2)^3 \times 50} = 100$

$\Delta G = \Delta G^o + RT\ell nQ$

$= -31.06 + \dfrac{2500}{1000} \times 4.6$

$= -31.06 + 2.5 \times 4.6$

$= -31.06 + 11.5$

$= -19.56$

$\Delta G = \Delta H - T\Delta S$

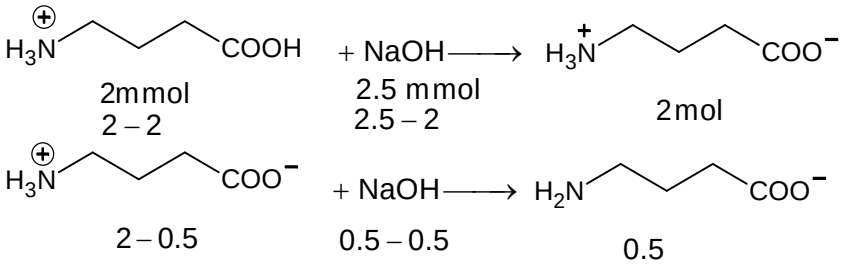
$-19.56 = -4.2 - T\Delta S$

$-15.36 = -T\Delta S$

$\Delta S = \dfrac{15.36}{300} \times 1000$ JK^{-1}

$= 51.2$ JK^{-1}

31. 00010.22

Sol. 

$pH = pK_{a_2} + \log \dfrac{[C.B]}{[Acid]}$

$= 10.7 + \log \dfrac{0.5 / V}{15 / V}$

$$pH = 10.7 + \log\frac{1}{3}$$

$= 10.7 - 0.48$

$= 10.22$

32. 00001.25

Sol. $\Lambda^{\infty} = 2x + 4xx_2 - 3xx_2 = 4x$

$\left|\frac{k_1}{k_2} = \frac{G_1}{G_2} = \frac{R_2}{R_1}\right.$ k_1 for $Ba(OH)_2$ k_2 for KCl

$$\frac{k_1}{k_2} = \frac{1}{4000} \Rightarrow k_1 = \frac{k_2}{4000} = \frac{0.002x}{4000}$$

$$\Lambda_m^c = \frac{k_1 \times 1000}{M} = \frac{0.002x \times 1000}{4000 \times 10^{-2}} = \frac{0.2x}{4} = \frac{2x}{40} = \frac{x}{20}$$

$$\alpha = \frac{\Lambda_m^c}{\Lambda_M^{\infty}} = \frac{x}{20 \times 4x} = \frac{1}{8} \times 10^{-1} = 0.125 \times 10^{-1}$$

$$= 1.25 \times 10^{-2}$$

33. 00013.20

Sol.

$$A(g) \longrightarrow 2B(g) + C(s) + D(g)$$

	P_o		
10 min	$P_o - P$	$2P$	P
	$P_o - P_o$	$2P_o$	P_o

$3P_o = 27$

$P_o = 9\,bar$

$P_T = P_o - P + 2P + P$

$P_T = P_o + 2P$

$21 = 9 + 2P$

$2P = 12$

$P = 6atm$

$$k = \frac{1 \times 60}{10} \ln\frac{9}{9-6}$$

$= 6\ell n3$

$= 6 \times 1.1$

$= 6.6\ min^{-1}$

$r = k \times [A]$

$= 6.6 \times 2$

$=13.2$ mole/litre min^{-1}

34. 00000.82

Sol. $AgBr(s) \rightleftharpoons Ag^+ + Br^-$ $\quad \Delta G_1^o = -RT\ell nK_{SP}$

$Ag^+ + e^- \rightleftharpoons Ag$ $\quad \Delta G_2^o = -nFE_1^o$

$AgBr + e^- \rightleftharpoons Ag + Br^-$ $\quad \Delta G_3^o = -nFE_2^o$

$\Delta G_3^o = \Delta G_1^o + \Delta G_2^o$

$-1FE_2^o = -1FE_1^o - RT\ell nK_{SP}$

$$E_2^o = E_1^o + \frac{RT}{F}\ell nK_{SP}$$

$0.0725 = E_1^o + 0.025(-29.9)$

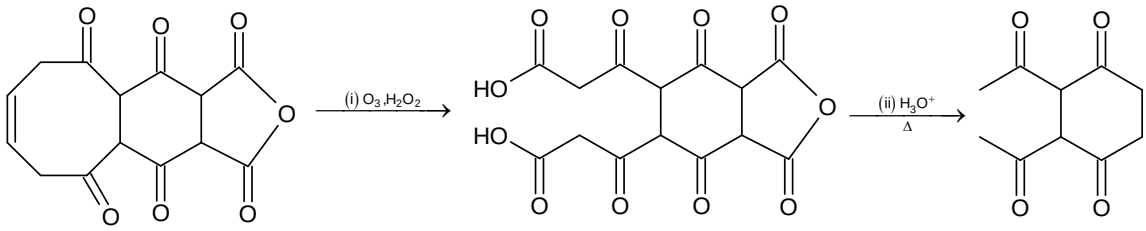
$0.0725 = E_1^o - 0.7475$

$E_1^o = 0.0725 + 0.7475$

= 0.82

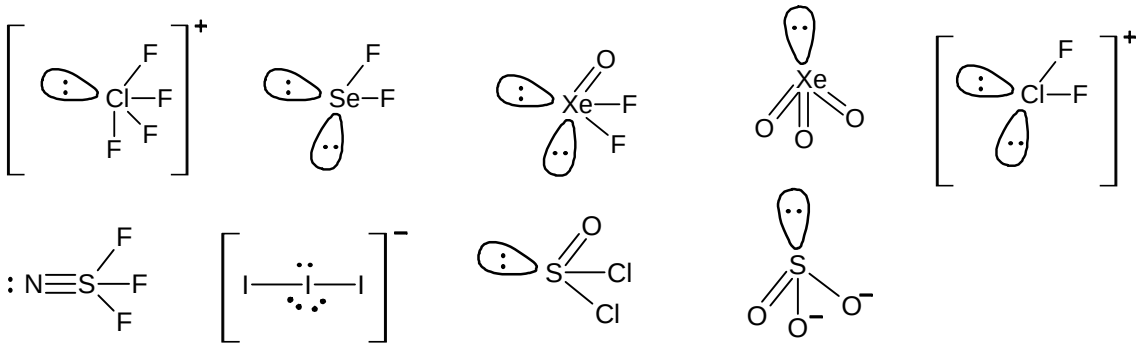
35. 00000.80

Sol.



36. 00001.50

Sol.



Mathematics PART – III

SECTION – A

37. B, C, D

Sol. Consider $\Delta A'B'C'$, where $\angle A' = \frac{2\pi}{3} - \angle A$, $\angle B' = \frac{2\pi}{3} - \angle B$, $\angle C' = \frac{2\pi}{3} - \angle C$

Now apply triangle inequality to get $S > -\frac{1}{\sqrt{3}}$

38. A, B, C

Sol. Take base as the x – y plane with one vertex as origin. A, B be vertices along the x, y-axes respectively

$\overrightarrow{OA} = 2\hat{i}$, $\overrightarrow{OB} = 2\hat{j}$, $\overrightarrow{OC} = \hat{i} + \hat{j} + 2\hat{k}$, where C is the apex

$\therefore\ \overrightarrow{OA} \times \overrightarrow{OB} = 4\hat{k}$; $\overrightarrow{OA} \times \overrightarrow{OC} = 2\hat{k} - 4\hat{j}$; $\overrightarrow{OB} \times \overrightarrow{OC} = -2\hat{k} + 4\hat{i}$

Volume is $\frac{1}{3} \times$ parallelopiped = $\frac{8}{3}$

Tetrahedron with half of pyramid's base and the pyramid's apex has volume = $\frac{4}{3}$

39. C, D

Sol. $F(x) = \int_1^n \frac{x^u}{u} du$; $u \in [1, n] \Rightarrow \frac{x^u}{u}$ is strictly monotonic in x

$\Rightarrow$ F(x) is strictly monotonic

$x\int_1^n \frac{du}{u} \le \int_1^n \frac{x^u}{u} du \le x^n \int_1^n \frac{du}{u}$ as $x \to 1^+$ and $x\int_1^n \frac{du}{u} \ge \int_1^n \frac{x^u}{u} du \ge x^n \int_1^n \frac{du}{u}$ as $x \to 1^-$

$\Rightarrow \lim_{x\to 1} F(x) = \ln n$

40. A, B

Sol. $\int_0^{n+1} f(x)dx - \int_0^n f(x)dx = \int_n^{n+1} n^{x-n} dx = \frac{1}{n^n}\left[\frac{n^x}{\ln n}\right]_n^{n+1} = \frac{n-1}{\ln n}$

$\int_0^{n+1} g(x)dx - \int_0^n g(x)dx = \int_n^{n+1} (x-n)^n dx = \frac{(x-n)^{n+1}}{n+1}\Big]_n^{n+1} = \frac{1}{n+1}$

$t \in (0, 1) \Rightarrow \int_0^t f(x)dx = 0$, $t \in \left[1, \frac{\pi}{2}\right) \Rightarrow \int_0^t f(x)dx < \frac{\pi}{2} - 1$

$t = 1 \Rightarrow \int_0^t g(x)dx = 1$

$\lim_{n\to\infty} \int_0^n g(x)dx = \lim_{n\to\infty} \sum_{i=0}^n \frac{1}{i+1} \to$ Divergent series

41. A, B, D

Sol. For $x \in \left(0, \frac{\pi}{2}\right)$, f(x) = 0

For $x \in \left(\frac{\pi}{2}, \pi\right)$, $f(x) = \pi - x - x = \pi - 2x < 0$

For $x \in \left(\pi, \frac{3\pi}{2}\right)$, $f(x) = \pi - x - (2\pi - x) = -\pi < 0$

for $x \in \left(-\pi, \frac{-\pi}{2}\right)$, $f(x) = -(\pi + x) - (-x) = -\pi < 0$

42. D

Sol. Incident ray, if extended, passes through one focus (2, 0). So reflected ray will follow the path to pass through the other focus (–2, 0)

43. A

Sol. Area $= 2 \cdot \frac{1}{2}\int_0^{\frac{\pi}{4}} \frac{9\cos^2\theta\sin^2\theta}{(\cos^3\theta + \sin^3\theta)^2} d\theta = \frac{3}{2}$

44. C

Sol. Length $= \int_0^1 \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = \int_0^1 \sqrt{\theta^2 + 1} d\theta = \left[\frac{\theta}{2}\sqrt{1+\theta^2} + \frac{1}{2}\ln\left|\theta + \sqrt{1+\theta^2}\right|\right]_0^1 = \frac{1}{2}\ln\left(1+\sqrt{2}\right) + \frac{1}{\sqrt{2}}$

45. A

46. C

Sol. (for Q.45.-46)

Let A(h, k), C(x_0, y_0), B(x_1, y_1)

$h + x_0 = 9(k + y_0)$ and $h^2 + k^2 = x_0^2 + y_0^2 \Rightarrow x_0 = \frac{1}{41}(40h + 9k)$, $y_0 = \frac{1}{41}(9h - 40k)$

$h + x_1 = -(k + y_1)$ and $h^2 + k^2 = x_1^2 + y_1^2 \Rightarrow x_1 = -k, y_1 = -h$

$\frac{5 - y_1}{4 - x_1} = \frac{5 - y_0}{4 - x_0} \Rightarrow 4h^2 + 4k^2 + 41k = 0$

SECTION – D

47. 00001.00

Sol. Two bishops attack each other if they are in squares along the same diagonal. The number of ways of choosing two squares along the same diagonal on a 8×8 chessboard is

$2 \times \left(2\left({}^2C_2 + {}^3C_2 + ... + {}^7C_2\right) + {}^8C_2\right) = 2\left({}^8C_3 + {}^9C_3\right)$

$\therefore P = 1 - \frac{2\left({}^8C_3 + {}^9C_3\right)}{{}^{64}C_2} = 1 - \frac{280 \times 2}{64 \times 63} \Rightarrow \left[\frac{1}{P}\right] = 1$

48. 00000.20

Sol. $\begin{vmatrix} x & x^2 & 1-yz \\ y & y^2 & 1-zx \\ z & z^2 & 1-xy \end{vmatrix} = (x-y)(y-z)(z-x)(1-xy-yz-zx)$

49. 00002.00

Sol. $|\text{adj } \lambda A| = |\lambda A|^2 = \lambda^6|A|^2 = \lambda^6$ (as A is orthogonal)

$|\text{Adj}(\text{adj } \lambda A)| = |\text{adj } \lambda A|^2 = \lambda^{12}$

$\therefore \frac{\lambda^{12}}{\lambda^6} = \lambda^6 = 16 \Rightarrow [\lambda^2| = \left[\sqrt[3]{16}\right] = 2$

50. 00003.00

Sol. $\int_0^1 \lim_{n\to\infty} \sum_{k=0}^{n} \frac{x^{k+1}3^k}{k!}dx = \int_0^1 x \lim_{n\to\infty} \sum_{k=0}^{n} \frac{x^k 3^k}{k!}dx = \int_0^1 xe^{3x}dx = \frac{xe^{3x}}{3}\Big|_0^1 - \frac{e^{3x}}{9}\Big|_0^1 = \frac{2e^3+1}{9}$

51. 00000.60

Sol. Probability that (i) Dice shows 2 and A reports it correctly, $\frac{1}{6}\times\frac{1}{3} = \frac{1}{18}$

(ii) Dice did not show 2 and A reported it as 2, $\frac{5}{6}\times\frac{2}{3}\times\frac{1}{5} = \frac{1}{9}$

(iii) Dice showed 4 or 6 and A reported it as 2, $\frac{2}{6}\times\frac{2}{3}\times\frac{1}{5} = \frac{2}{45}$

$$P = \frac{\frac{1}{18}+\frac{2}{45}}{\frac{1}{18}+\frac{1}{9}} = \frac{3}{5}$$

52. 00000.50

Sol. $\frac{1}{{}^nC_0}+\frac{1}{{}^nC_n} \le a_n \le \left(\frac{1}{{}^nC_0}+\frac{1}{{}^nC_n}\right)+\left(\frac{1}{{}^nC_1}+\frac{1}{{}^nC_{n-1}}\right)+\sum_{i=2}^{n-2}\frac{1}{{}^nC_2}$

53. 00000.02

Sol. $T_k = \frac{51!}{50}\left[\frac{1}{(k+1).....(k+50)} - \frac{1}{(k+2)(k+3).....(k+51)}\right]$

$S = \frac{1}{50}$

54. 00000.05

Sol. Other 2 vertices lie on the director circle

L = $2\sqrt{5}$